

Melvin Orichi Socana

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EDUCATION

Stanford University, Stanford, CA, GPA 3.53 September 2022 – May 2024
Master of Science (MS): Electrical Engineering (Hardware & Software)

North Carolina State University, Raleigh, NC, GPA 3.43 August 2018 – May 2022
Bachelor of Science (BS): Electrical & Computer Engineering

EXPERIENCE

Amazon, Seattle, WA May 2023 – August 2023

Software Development Engineer Intern

- Spearheaded the development of 'Albums' API feature for Amazon Photos software in C# and .NET framework.
- Authored a design document under ambiguous requirements, aligning cross-functional teams on architecture.
- Communicated effectively with cross-functional teams and non-technical stakeholders.
- Engineered SQL database queries and schemas to efficiently store/retrieve metadata via pagination.
- Collaborated with senior engineers to integrate backend logic with existing RESTful API endpoints.

Intelligent Wireless Networking Laboratory, Raleigh, NC February 2022 – May 2022

Software/Artificial Intelligence & Wireless Networking Research Assistant

- Collaborated with Cisco & PhD candidate to troubleshoot and optimize network performance and throughput.
- Deployed lightweight machine learning models on resource-constrained nodes, demonstrating the practicality of AI-driven solutions in low-power and limited-hardware environments.

Active Robotic Sensing Laboratory, Raleigh, NC August 2021 – May 2022

Software, Robotics, & Machine Learning Research Assistant

- Led development on the imaging system for the BenchBot, a robotics device designed to identify species and estimate biomass.
- Designed and implemented a robust script to interface with DepthAI cameras and servos, capturing high-quality plant images to train machine learning models.

Johnson & Johnson, Santa Clara, CA May 2021 – August 2021

Software/Hardware Engineer Intern

- Engineered a diagnostic board for rapid verification of communication protocols, yielding 20-30% cost savings.
- Authored functional testing protocols and reports for complex electronic assemblies.

RESEARCH

Sequence-to-Sequence Semantic Parsing over Wikidata | *Python, Machine Learning, NLP*
<https://github.com/farhadGithub/ReFinED-WithGPT> | <https://arxiv.org/abs/2305.14202>

- Developed a few-shot sequence-to-sequence semantic parser with RAG, enabling natural language queries to generate accurate SPARQL queries. Enhanced entity-linking pipeline by 20%, leading to increased performance.

TECHNICAL SKILLS

Programming Languages
Software & Tools
Frameworks & Libraries
Cloud & DevOps

Python, C, C++, C#, Go, JavaScript, TypeScript, MATLAB, SQL
Linux, SQL, NoSQL, Docker, KiCad, Git, Artificial Intelligence
.NET Core, PyTorch, NodeJS, React, Selenium, NumPy
AWS, GCP, Kubernetes, REST API, TCP/IP, CI/CD